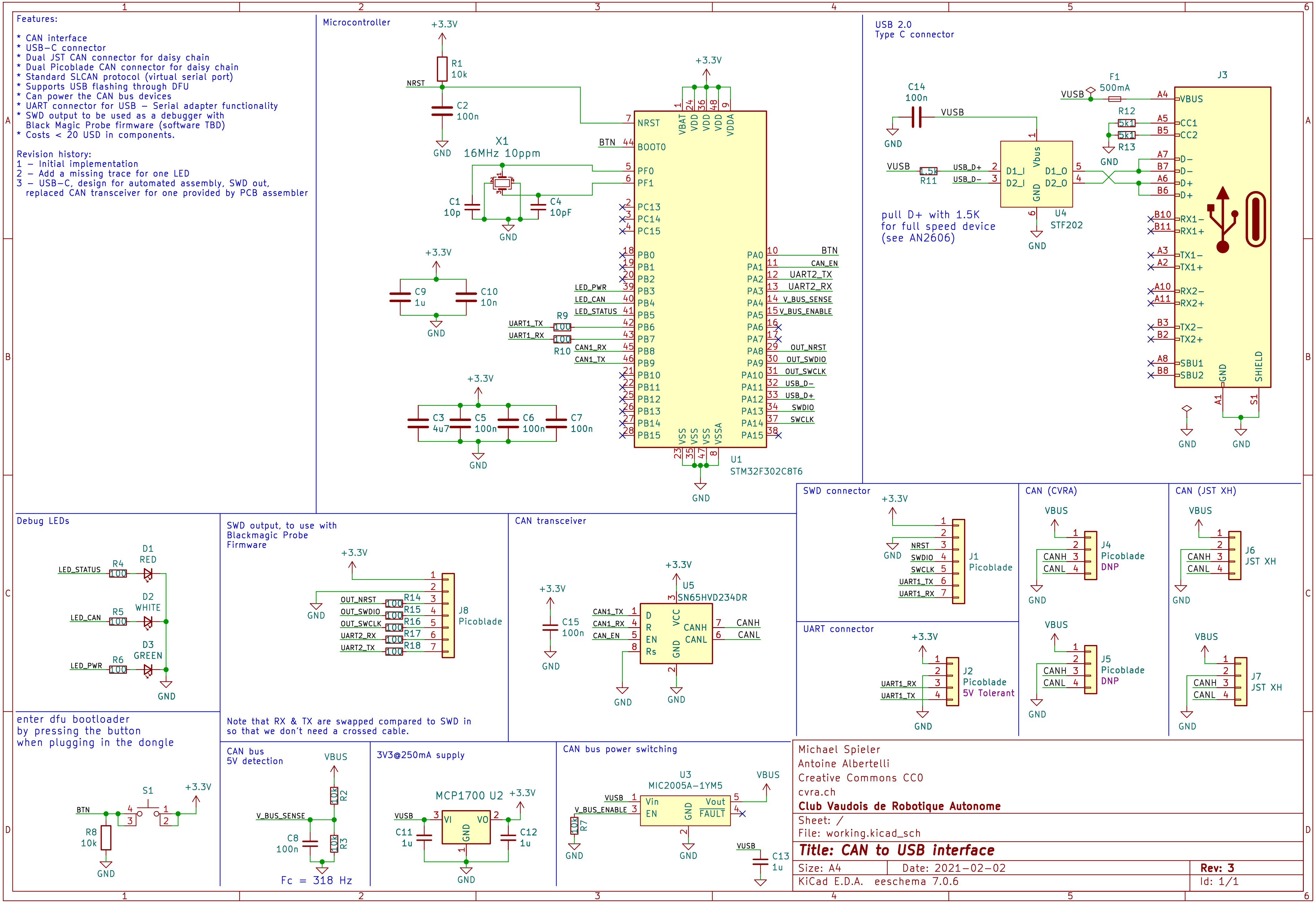
Features:
* CAN interface
* USB−C connector
* Dual JST CAN connector for daisy chain
* Dual Picoblade CAN connector for daisy chain
* Standard SLCAN protocol (virtual serial port)
* Supports USB flashing through DFU
* Can power the CAN bus devices
* UART connector for USB − Serial adapter functionality
* SWD output to be used as a debugger with Black Magic Probe firmware (software TBD)
* Costs < 20 USD in components.
Revision history:
1 − Initial implementation
2 − Add a missing trace for one LED
3 − USB−C, design for automated assembly, SWD out, replaced CAN transceiver for one provided by PCB assembler
Microcontroller
X1 16MHz 10ppm
U1 STM32F302C8T6
USB 2.0 Type C connector
pull D+ with 1.5K for full speed device (see AN2606)
U4 STF202
J3
F1 500mA
Debug LEDs
D1 RED
D2 WHITE
D3 GREEN
SWD output, to use with Blackmagic Probe Firmware
J8 Picoblade
Note that RX & TX are swapped compared to SWD in so that we don't need a crossed cable.
CAN transceiver
U5 SN65HVD234DR
SWD connector
J1 Picoblade
UART connector
J2 Picoblade 5V Tolerant
CAN (CVRA)
J4 Picoblade DNP
J5 Picoblade DNP
CAN (JST XH)
J6 JST XH
J7 JST XH
enter dfu bootloader by pressing the button when plugging in the dongle
CAN bus 5V detection
Fc = 318 Hz
3V3@250mA supply
MCP1700 U2
CAN bus power switching
U3 MIC2005A−1YM5
Michael Spieler
Antoine Albertelli
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cvra.ch
Club Vaudois de Robotique Autonome
Sheet: /
File: working.kicad_sch
Title: CAN to USB interface
Size: A4
Date: 2021−02−02
Rev: 3
KiCad E.D.A. eeschema 7.0.6
Id: 1/1